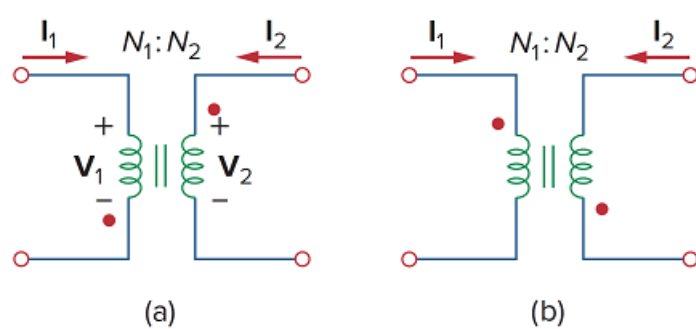


Problem

For the ideal transformer in Fig. 13.70(b), $N_2/N_1 = 10$. The ratio I_2/I_1 is:

- (a) 10
- (b) 0.1
- (c) -0.1
- (d) -10

Figure 13.70



Step-by-step solution

Step 1 of 2

Refer to Figure 13.70 (b) in the textbook.

From the Figure, the current, I_1 is entering to dotted terminal and the current, I_2 is leaving the dotted terminal.

Therefore, the ratio of primary to secondary current is $+n$.

Step 2 of 2

Calculate the ratio, $\frac{I_2}{I_1}$.

$$\begin{aligned}\frac{I_2}{I_1} &= +\frac{1}{n} \\ &= \frac{1}{\left(\frac{N_2}{N_1}\right)}\end{aligned}$$

Substitute 10 for $\frac{N_2}{N_1}$.

$$\begin{aligned}\frac{I_2}{I_1} &= \frac{1}{10} \\ &= 0.1\end{aligned}$$

Thus, the correct answer is **(b)**.